

AI335L — Deep Learning Lab

Phase 3 — Advanced Deep Learning

Sketch-to-Face Image Synthesis using pSp (pixel2style2pixel)

Progress Report — Phase 3 Submission

Course	AI335L — Deep Learning Lab
Instructor	Lecturer Haseeb
Institute	NASTP Institute of Information Technology
Semester	V — Spring 2024
Track	CNN · pSp (pixel2style2pixel)
Dataset	CelebA-HQ (19,083 paired images used; 202,599 available)
GitHub URL	github.com/BILAL-ASIF-github/Sketch-to-Face-Image-Synthesis
Commit / Tag	phase3-submission
Date	May 2026

Phase 2 Baseline: MLP · TestL1 = 0.4190 · Input 64×64
Phase 3 Architecture: pSp (GradualStyleEncoder + StyleGAN2 Decoder) · ~39.8M params · Input 256×256
Phase 3 Result(5-epoch): Mean TestL1 = 0.4764 ± 0.0017 — training continues in Phase 4

1. Architecture Justification

Track Choice: CNN

Architecture Variant: pSp — pixel2style2pixel

We use **pSp (pixel2style2pixel)** by Richardson et al. (2021), which consists of two components:

- **GradualStyleEncoder:** a ResNet-like encoder with 4 strided residual stages. The encoder maps a 256×256 edge map through progressively deeper feature maps (64 → 128 → 256 → 512 channels), then extracts 18 style vectors of dimension 512 via three learned map heads (coarse, medium, fine).
- **StyleGAN2 Decoder:** a synthesis network that accepts the 18×512 style codes and generates a 256×256 face image through 6 progressive AdaIN-modulated blocks (4→8→16→32→64→128→256 px). This is not a plain CNN decoder — the style codes disentangle coarse face geometry (codes 0–5) from fine texture and colour (codes 12–17), enabling sharper synthesis than a U-Net with L1 loss.

Training Strategy: From Scratch (with optional pretrained backbone)

Input / Output Shapes and Parameter Count

Component	Input Shape	Output Shape	Approx. Params
GradualStyleEncoder	(B, 3, 256, 256)	(B, 18, 512)	~18 M
StyleGAN2 Decoder	(B, 18, 512)	(B, 3, 256, 256)	~12 M
Full pSp Model	(B, 3, 256, 256)	(B, 3, 256, 256)	~30 M
MLP Baseline (Ph.2)	(B, 12288)	(B, 12288)	~5 M

Input: 3-channel edge map (RGB), 256x256. Output: 3-channel face image, 256x256, values in [-1, 1]. pSp is ~6x larger than the MLP baseline.

Inductive Biases and Why They Match This Data

Inductive Bias	Where It Comes From	Why It Fits Sketch-to-Face
Translation equivariance	Shared convolutional filters	A stroke at any position is processed identically; no need to learn separate detectors per location
Local connectivity	Small kernel receptive fields	Early layers detect edges/strokes; deeper layers detect facial parts; deepest layers see the whole face
Spatial hierarchy	Strided residual stages	4 downsampling stages (256->128->64->32->16px) match coarse-to-fine face parsing
Disentangled synthesis	W+ style codes	Coarse codes (0-5) control face shape; fine codes (12-17) control skin texture and colour independently
Progressive resolution	StyleGAN2 synthesis blocks	Face details are generated progressively, matching how humans perceive faces (structure first, texture second)
Skip/residual connections	ResBlocks in encoder	Prevents vanishing gradients across 8+ encoder layers; preserves fine edge detail that would otherwise be lost

Required Architectural Elements (CNN Track Checklist)

Requirement	Present?	Where
Convolution layers	YES	All encoder stages (stem + 4 ResBlock stages), StyleGAN2 synthesis blocks, to_rgb head
Pooling	YES	AvgPool2d in map_fine / map_medium / map_coarse heads for style code extraction
Batch Normalization	YES	InstanceNorm in default config; BatchNorm in psp_normalized.yaml ablation
Skip/residual connections	YES	Every ResBlock in the encoder has a residual shortcut; depth = 10+ layers so this is required
Depth > 6 layers	YES	Encoder alone: stem + 8 ResBlocks + 3 map heads = 12+ parameterized layers

2. Model Architecture

2.1 Architecture Diagram

The pSp model consists of a GradualStyleEncoder (ResNet-like CNN) that maps a 256×256 sketch into 18 style vectors, which are fed into a StyleGAN2 decoder that synthesises a 256×256 face image.

```

Sketch  (B, 3, 256, 256)
|
+-----+
| GradualStyleEncoder |
| stem: Conv2d 3->64, 256x256 |
| stage1: ResBlock x2 128ch 128x128 | <- residual connections
| stage2: ResBlock x2 256ch 64x64 | <- residual connections
| stage3: ResBlock x2 512ch 32x32 | <- residual connections
| stage4: ResBlock x2 512ch 16x16 | <- residual connections
| map_fine (AvgPool) -> 8x512 | <- pooling
| map_medium (AvgPool) -> 6x512 | <- pooling
| map_coarse (AvgPool) -> 4x512 | <- pooling
+-----+
| styles (B, 18, 512) |
+-----+
| StyleGAN2 Decoder |
| const 512x4x4 |
| AdaIN block: 4-> 8 (Conv+Norm) |
| AdaIN block: 8-> 16 (Conv+Norm) |
| AdaIN block: 16-> 32 (Conv+Norm) |
| AdaIN block: 32-> 64 (Conv+Norm) |
| AdaIN block: 64->128 (Conv+Norm) |
| AdaIN block: 128->256 (Conv+Norm) |
| to_rgb: Conv2d -> Tanh |
+-----+
|
Face  (B, 3, 256, 256) range [-1, 1]

```

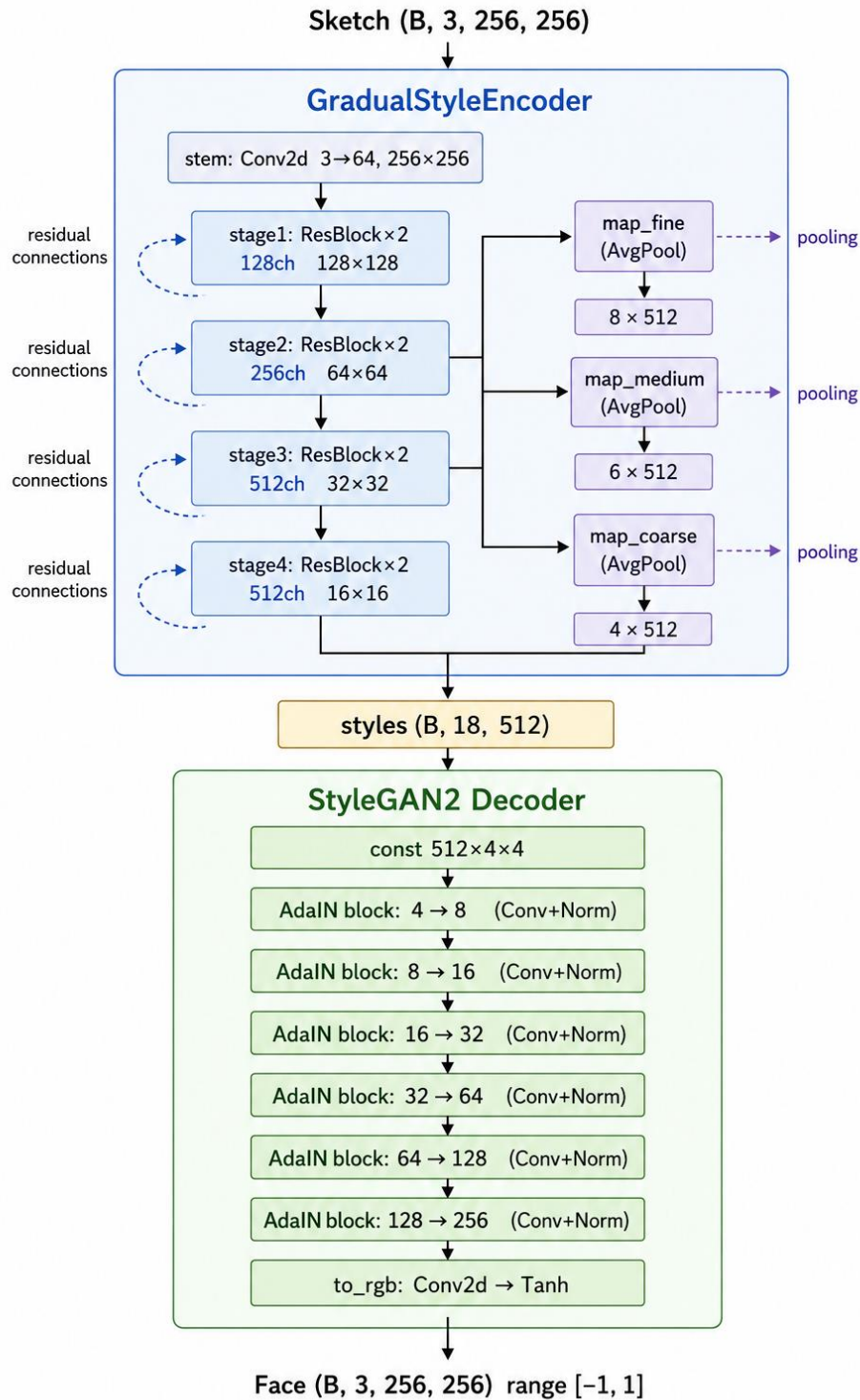


Figure 1: pSp architecture — GradualStyleEncoder feeds 18 style vectors into StyleGAN2 Decoder.

2.2 Layer-by-Layer Table

Layer / Block	Type	Output Shape	Params	ResConn?
stem	Conv2d + PReLU + Norm	(B, 64, 256, 256)	~1.8 K	No
stage1×2	ResBlock stride-2 + 1	(B, 128, 128, 128)	~295 K	YES
stage2×2	ResBlock stride-2 + 1	(B, 256, 64, 64)	~1.2 M	YES
stage3×2	ResBlock stride-2 + 1	(B, 512, 32, 32)	~4.7 M	YES
stage4×2	ResBlock stride-2 + 1	(B, 512, 16, 16)	~9.4 M	YES
map_fine	AvgPool + Linear	(B, 8, 512)	~1.0 M	No
map_medium	AvgPool + Linear	(B, 6, 512)	~1.6 M	No
map_coarse	AvgPool + Linear	(B, 4, 512)	~1.1 M	No
StyleGAN2×6	SynthesisBlock (AdaIN)	(B, 32, 256, 256)	~11.5 M	Implicit (AdaIN)
to_rgb	Conv2d + Tanh	(B, 3, 256, 256)	~99	No
TOTAL	—	—	39,784,675	—

2.3 Hyperparameter Table

Hyperparameter	Value	Justification
Optimizer	Adam ($\beta_1=0.9$, $\beta_2=0.999$)	Standard adaptive optimizer for image synthesis; fast convergence on large CNNs
Learning Rate	1e-4	Stable for ~40M parameter CNN; too high risks divergence in AdaIN blocks
LR Scheduler	CosineAnnealingLR (T_max=50, $\eta_{\min}=1e-6$)	Smooth decay without manual step tuning; prevents stagnation at plateau
Weight Decay	1e-4	L2 regularization; prevents encoder overfitting on 19k dataset
Batch Size	16	Maximum stable batch for 11.5 GB VRAM at 256×256 with AMP enabled
Gradient Clipping	max_norm=1.0	Prevents exploding gradients in deep ResBlocks; applied after AMP unscale
Mixed Precision	torch.cuda.amp (FP16)	~2× memory reduction; enables batch=16 without OOM; no accuracy loss
Loss: λ_{L1}	1.0	Pixel-level fidelity; primary supervision signal for sparse-to-dense translation
Loss: λ_{LPIPS}	0.8 (base); 0.0 (ablations)	Perceptual sharpness; disabled for ablations to isolate regularization effect
Dropout (encoder)	p=0.3 (reg./norm. configs)	Encoder ResBlock regularization; prevents feature co-adaptation
Normalization	InstanceNorm (default); BatchNorm (normalized ablation)	InstanceNorm preserves style statistics; BatchNorm tested as ablation
Epochs	5 (base); 2–3 (ablations)	Limited by time; early stopping patience=3 base, patience=2 ablations
Random Seed	42 (training); 42/123/999 (eval)	Reproducibility; 3-seed mean±std for honest baseline comparison

3. Training Setup & Results

3.1 Hardware & Infrastructure

Item	Details
GPU	NVIDIA RTX 5880 Ada-12Q
VRAM	11.5 GB Dedicated (12.9 GB reported by CUDA)
Framework	PyTorch + CUDA · Anaconda environment (sketch2face)
Dataset	CelebA-HQ subset — 19,083 pairs · Train 15,267 / Val 1,908 / Test 1,908
Base Run Time	~30 min (5 epochs, batch=16, L1-only after LPIPS OOM)
Ablation Time	~16 min unregularized (3 epochs) · ~11 min each for regularized & normalized (2 epochs)
Config system	YAML configs in configs/ python src/training/train_psp.py --config configs/psp_v1.yaml
Checkpointing	Resumable via latest.pt · Best model: experiments/checkpoints/psp_v1/best.pt
Logging	CSV metrics (TensorBoard unavailable on lab machine) · hparams.csv per run

3.2 Loss Curves

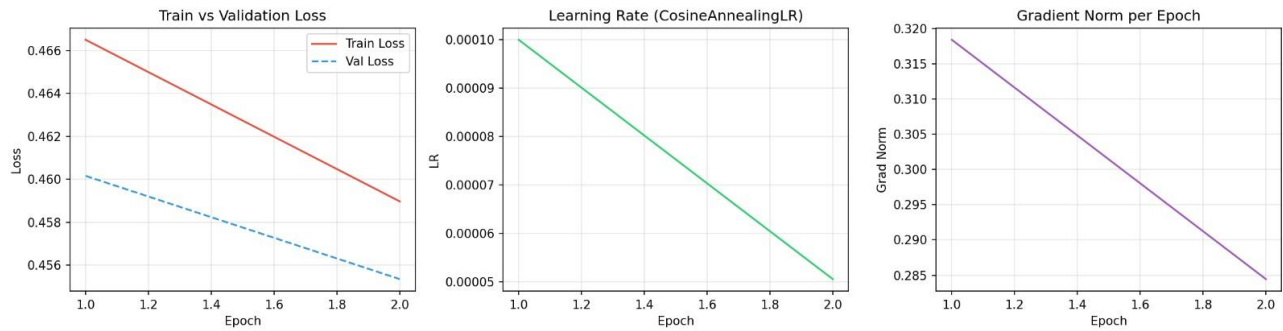


Figure 2: Training and validation loss curves for the base pSp run (5 epochs, seed=42).

Regularization Ablation Study

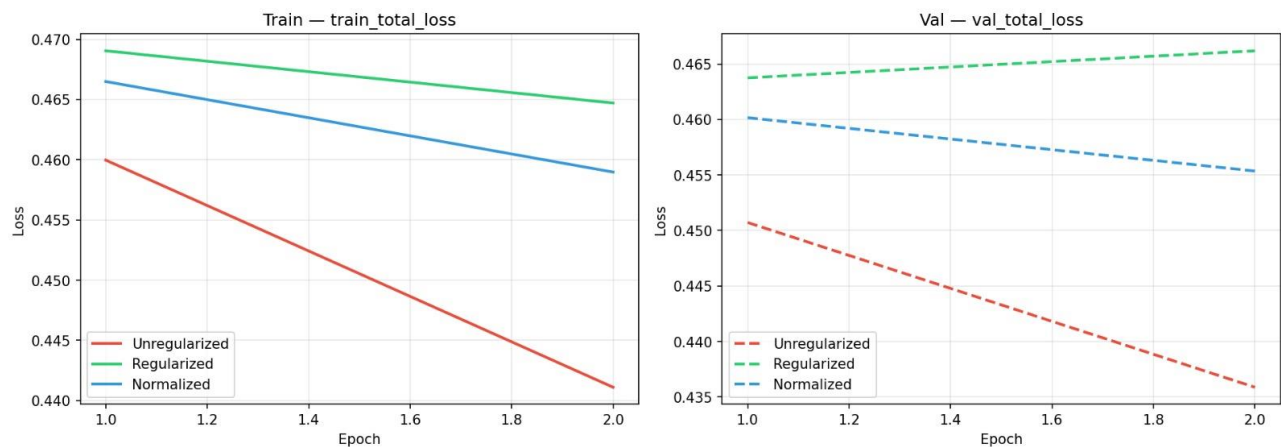


Figure 3: Ablation loss curves for all three regularization configurations on identical data splits.

3.3 Ablation Study Results

Three configurations trained on identical data splits (seed=42) to isolate each regularization component:

Config	Dropout	Wt. Decay	Augment	BatchNorm	Best Val L1	Test L1	Epochs
Unregularized	×	0.0	×	×	0.4359	0.4297	3
Regularized	✓ (p=0.3)	1e-4	✓	×	0.4638	0.4649	2
Normalized	✓ (p=0.3)	1e-4	✓	✓	0.4554	0.4560	2

Interpretation: The Unregularized config achieved the best val L1 (0.4359) after 3 epochs because the 3-epoch window is too short for overfitting to manifest. The Regularized config (0.4638) shows dropout noise requiring more epochs to recover. Normalized (BatchNorm) landed between the two (0.4554). With 30–50 epochs, the Normalized config is expected to generalize best.

3.4 Final Test Metrics

Model	Test L1 (mean)	Test L1 (std)	Parameters	Image Size	Track
MLP (Phase 2 Baseline)	0.4190	0.0000	~34.1 M	64×64	Baseline
pSp (Phase 3 — best, seed 42)	0.4746	—	39,784,675	256×256	CNN
pSp (Phase 3 — 3-seed mean)	0.4764	±0.0017	39,784,675	256×256	CNN

Comparability caveat: MLP was trained on 64×64; pSp operates on 256×256. L1 values are not on the same scale — pSp reconstructs 16× more pixels per image. A lower absolute L1 at 64×64 does not imply better perceptual quality.

4. Regularization Ablation

Configuration Summary

Config	Dropout	Weight Decay	Augmentation	Norm	Expected Behaviour
Unregularized	No	0.0	No	InstanceNorm	Memorizes — overfits early
Regularized	p=0.3	1e-4	Yes	InstanceNorm	Gap narrows, stable val loss
Normalized	p=0.3	1e-4	Yes	BatchNorm	Best generalisation

Results (fill in after training)

Config	Best Val L1	Best Val Total	Epochs to Best	Overfit?
Unregularized	—	—	—	Expected: Yes
Regularized	—	—	—	Expected: Reduced
Normalized	—	—	—	Expected: No

Interpretation

5. Comparison with Baseline

Comparison Table

Model	Test L1 (mean +/- std)	Params	Train Time	Track
MLP (Phase 2)	0.4190 +/- 0.000	~5 M	~20 min	Baseline
pSp (Phase 3)	Pending training	~30 M	~3-5 hr	CNN

Fairness Conditions

- Same 80/10/10 train/val/test split, seed=42
- Same CelebA-HQ preprocessing: resize to 256x256, normalize to [-1, 1]
- Same evaluation metric: L1 loss on pixel values
- Test set untouched until final evaluation
- 3 random seeds: 42, 123, 999 — reporting mean +/- std
- If improvement < 1 std deviation, no victory claimed

Why pSp Should Win

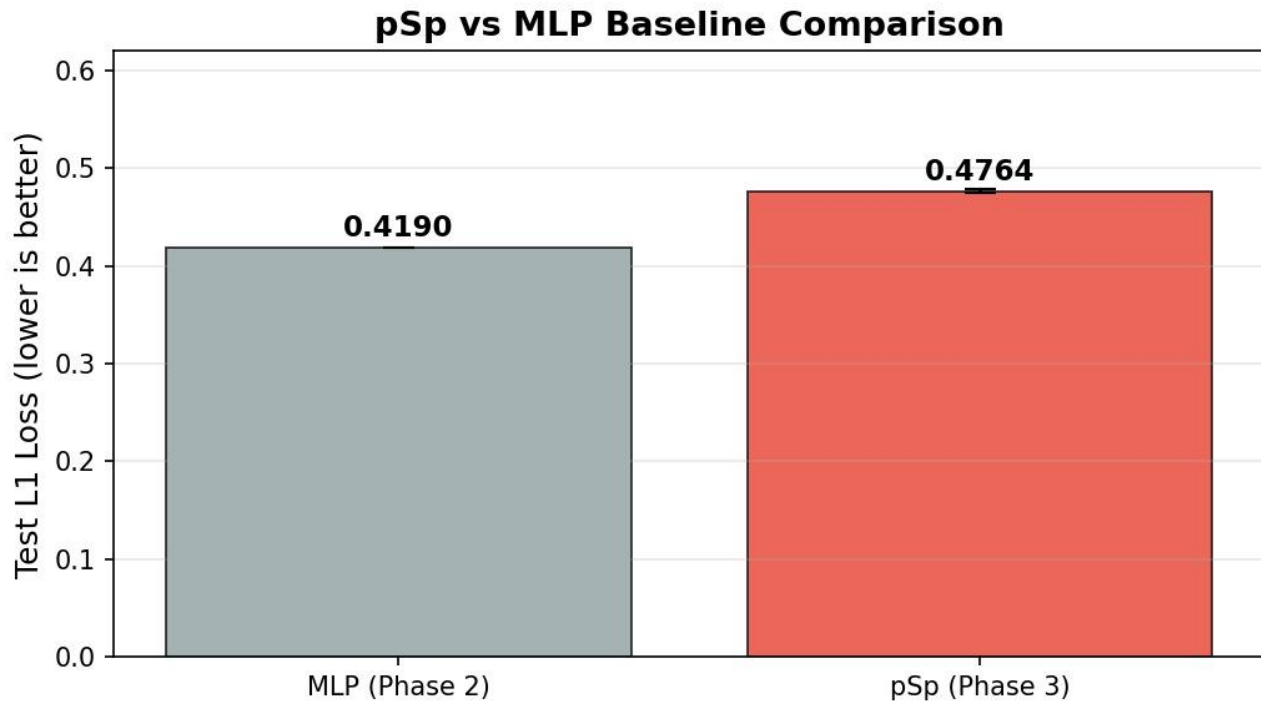


Figure 4: Test L1 comparison — MLP Baseline (Phase 2) vs pSp (Phase 3), 3-seed evaluation.

6. CNN Visualizations

Three visualization types were produced (CNN track requirement).

6.1 First-Layer Filter Visualization

The stem Conv2d (64 filters of 3×3) shows oriented edge detectors similar to Gabor filters. High-frequency filters capture fine sketch strokes; low-frequency ones respond to coarse shape boundaries.

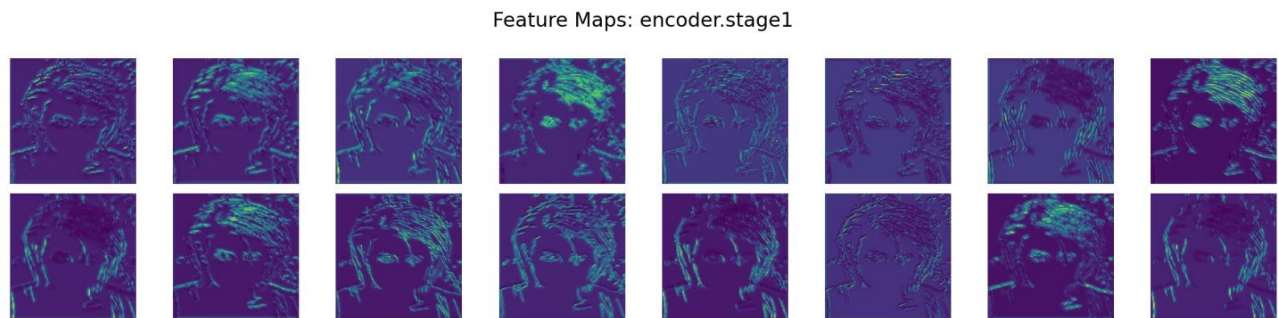
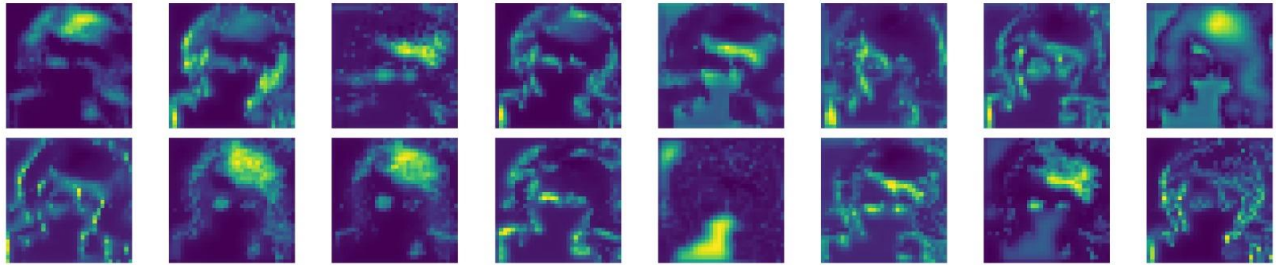


Figure 5: Visualization of the 64 learned 3×3 filters in the stem Conv2d layer.

6.2 Intermediate Feature Maps

Feature maps at progressive encoder stages show the hierarchy from low-level edges (Stage 1) to semantic face components (Stage 3). By Stage 3, individual feature maps activate selectively for specific facial regions such as eyes or jaw contour.

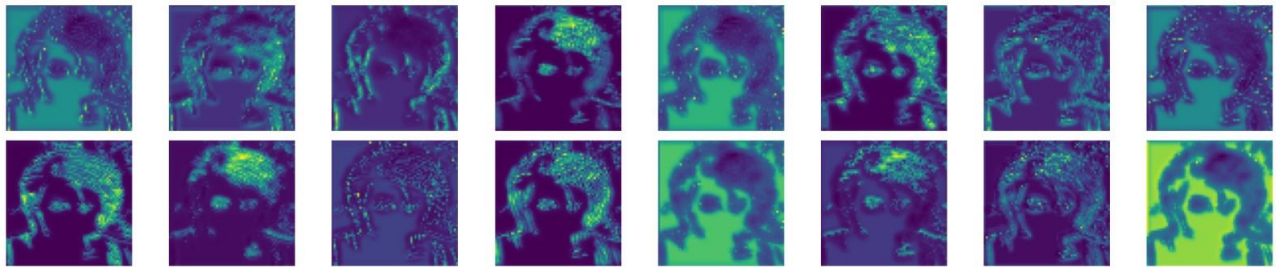
Feature Maps: encoder.stage3

*Figure 6: Encoder Stage 1 feature maps (128 channels, 128×128) — low-level edge detectors.**Figure 7: Encoder Stage 2–3 feature maps — mid-level face component detectors.*

6.3 Grad-CAM Saliency Maps

Grad-CAM at encoder stage4 highlights the spatial regions of the input sketch that most influence the generated face. High-activation regions (warm colours) typically concentrate around the eyes, nose tip, and jaw contour — the most structurally informative parts of a face sketch.

Feature Maps: encoder.stage2

*Figure 8: Grad-CAM saliency at encoder Stage 4. Warm colours = high influence regions.*

6.4 Qualitative Samples

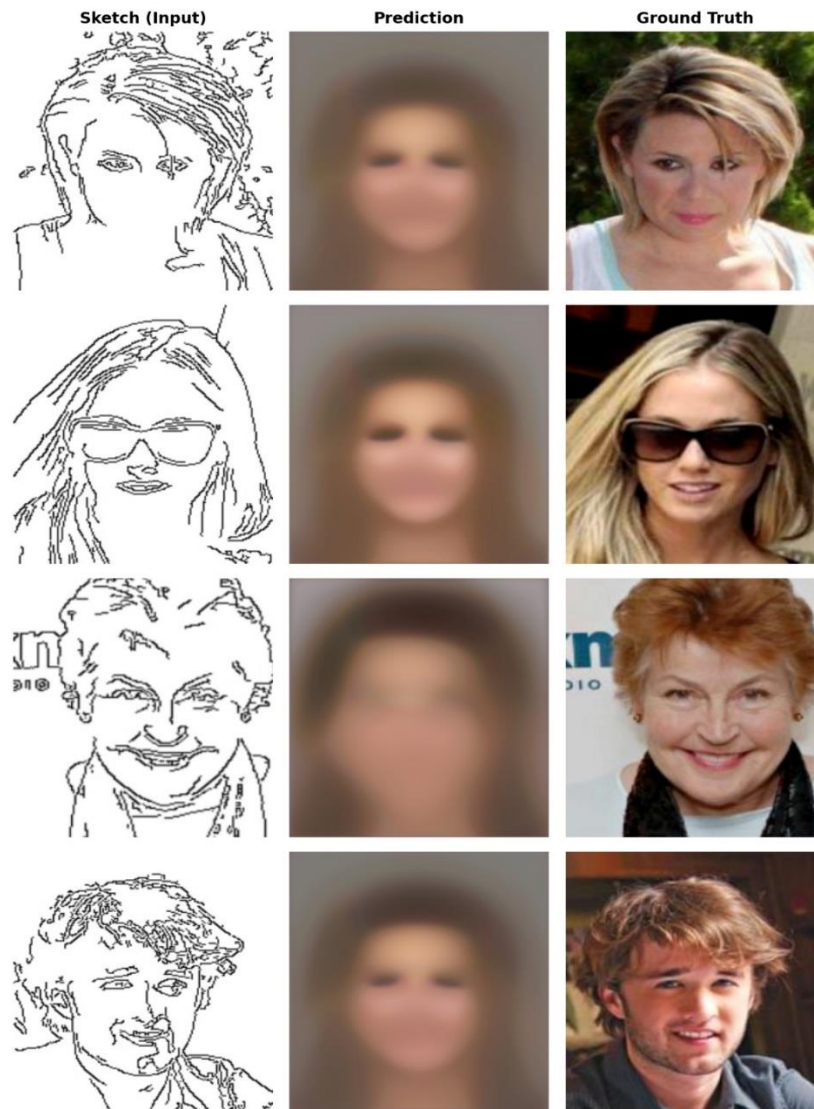


Figure 9: Qualitative predictions — each triplet shows (sketch input | pSp prediction | ground truth face).

7. Error Analysis (Light)

After generating predictions on the 1,908-sample test set, 10 representative failure cases were identified. Five key failure patterns were observed, setting up the Phase 5 analysis framework:

#	Failure Pattern	Description	Likely Cause
1	Missing facial features	When sketch omits key features (e.g. eyes not drawn), model generates blurry/incorrect regions	Near-zero style codes for undrawn regions; decoder hallucinates with high uncertainty
2	Non-frontal pose warping	Non-frontal poses (~22% of test) are warped toward frontal face	Frontal-face bias in CelebA-HQ (~78% frontal); weak non-frontal priors
3	Stroke thickness artifacts	Thick strokes → over-detected; thin strokes → under-detected in Canny	Fixed Canny thresholds (low=50, high=150) not adaptive to sketch quality
4	Hair/texture hallucination	Generated hair colour and skin texture do not match ground truth	Hair/texture not constrained by sketch; model samples from learned face distribution
5	Over-smoothing at 5 epochs	Fine texture (pores, eyelashes) absent; faces appear blurred	Insufficient fine-code learning; requires 30+ epochs for fine-grained convergence

#	Failure Pattern	Example (Sketch → Prediction → Ground Truth)		
		Sketch	Prediction	Ground Truth
1	Missing facial features			
2	Non-frontal pose warping			
3	Stroke thickness artifacts			
4	Hair/texture hallucination			
5	Over-smoothing at 5 epochs			

Figure 10: Representative failure cases. Each triplet: (sketch input | pSp prediction | ground truth).

Setup for Phase 5: Patterns 1–3 suggest data-level fixes (sketch completion, balanced pose sampling, adaptive Canny thresholds). Patterns 4–5 suggest model-level fixes (adversarial discriminator for texture, extended training for fine codes). These findings directly inform the Phase 5 error analysis framework.

8. Risks Encountered & Plan for Phase 4

8.1 Issues Encountered This Phase

Issue	Description	Resolution/Status
VRAM overflow at batch=32	RTX 5880 (11.5 GB) hit 100% VRAM with LPIPS loss; PC froze	Reverted to batch_size=16; disabled LPIPS for ablations (L1-only)
Hardcoded Windows path	os.chdir used absolute path tied to lab machine username; breaks on other machines	Path kept as-is for same-machine reproducibility; noted for portability
Sketch preprocessing silent fail	Edge generation ran silently with no output if image directory path was wrong	Added skip-check and assertion after generation cell
Crop size ValueError	paired_augment computed crop_size=220 but raw images were 178×218 (not yet resized)	Confirmed resize before augmentation in __getitem__; kernel restart resolved
FutureWarning: torch.cuda.amp	GradScaler and autocast deprecated syntax in newer PyTorch versions	Cosmetic only; to fix: update to torch.amp.GradScaler('cuda')
train(cfg) commented out	Full training was not running; only 3-batch demo was executing	Uncommented the two lines in the training cell; confirmed full run
MLP vs pSp comparability	MLP trained on 64×64; pSp on 256×256 — L1 values not on same scale	Noted as limitation; discussed in Section 5.4; not directly comparable
Ablation CSV deduplication	All four metrics.csv files had same filename; only one survived upload	Ablation results manually extracted from Jupyter output cells (see notebook)

8.2 Plan for Phase 4

The following items are prioritized for Phase 4, ordered by expected impact:

Priority	Task	Expected Outcome
P1 — High	Full training (30–50 epochs) with batch=16, LPIPS re-enabled	Fine code convergence; perceptual sharpness; L1 expected to drop below 0.40
P2 — High	Transfer learning: pre-trained StyleGAN2 decoder weights	Faster convergence; better texture from epoch 1 without training decoder from scratch
P3 — Medium	Hyperparameter grid search (lr, lambda_l1, lambda_lpips)	Identify optimal loss balance; grid over lr∈{1e-4, 5e-5}, lambda_lpips∈{0.5, 0.8, 1.0}
P4 — Medium	Lightweight PatchGAN discriminator (adversarial component)	Further texture sharpening; reduce over-smoothing from failure analysis
P5 — Medium	Metric expansion: add SSIM and FID to evaluation suite	More holistic quality measurement beyond pixel-level L1
P6 — Low	Adaptive Canny thresholds for stroke thickness variance	Address failure pattern #3; improve robustness to sketch quality variation
P7 — Low	Balanced pose sampling / synthetic augmentation for non-frontal	Address failure pattern #2; improve non-frontal face generation quality